Using CNNs and Machine Learning to Diagnose Skin Cancer Moles

Anika Sivasankar
Massachusetts Academy of Math and Science
STEM Project
Instructor: Kevin Crowthers, Ph.D.

Table of Contents:

Good Laboratory Practice
Record-Keeping Contract

I, Anika Sivasankar, commit to record keeping in accordance with Good Laboratory Practices.

- My experiments and records will be reproducible, traceable, and reliable.
- I will NOT write my notes on scraps of paper, post-it notes, or other disposable items. My notes will go directly into my laboratory notebook.
- My data will be recorded in real-time. If I cannot record data in real-time, I will record raw data as soon as physically possible.
- I will record both qualitative and quantitative observations in my laboratory notebook and laboratory reports.
- My laboratory notebook will include information on the materials and instruments utilized during experimentation.
- I will initial and date (with time) over the edge of any material that is taped or pasted into my laboratory notebook.
- I will provide a real-time record of any analysis I perform.
- I will use blue or black pen to make entries in my laboratory notebook. I will NOT use a pencil.
- I will define ALL abbreviations.
- If I make a mistake in my laboratory notebook, laboratory worksheets, or other written material, I will not obliterate or obscure the mistake. Instead, I will cross out the mistake using a single line. Any empty spaces in tables or partially used notebook pages will be crossed out using a single diagonal line.
- If I record information online (ex. In Google Drive), I will log in so that my contributions are traceable.
- I will initial and date each page in my notebook and the front of each laboratory report.

<u>Anika Sivasankar</u>

Printed name: Anika Sivasankar

____

Signature: Anika

____

Date: 09/14/24

A more detailed description of GLP is located here:
https://docs.google.com/document/d/1zeYoNSniKTc7MlBgTG1SEnhJiCK3UimCvTcKPQcyHGw/edit?usp=sharing

Logbook Etiquette

For research and engineering purposes, a logbook is considered a legal document and will help in providing documentation for the origin of ideas.

1- When adding something written in Pen- Blue or Black ~~not a Pencil~~ (and DO NOT USE WHITEOUT- ~~mistakes~~ can be corrected by adding the information above the crossed out material and adding your initials and date
2- Don't worry about neatness- it is a living document but **should be legible but understandable**
3- Page Numbers should be consecutive and located on the top corner of the page- outer edge
4- Do not remove pages
5- Put a line through empty space
6- Neat handwriting
7- **Make an entry every time you work on your project**
8- Make sure your entries are verified by a mentor/ teacher signature and your signature
9- Organize your Notebook: Format
A: Table of Contents
B: Brainstorming and Topic Ideas
C: Project Introduction: Topic, Phrase 1(Testable Question/Engineering Need/Mathematical Conjecture), Phrase 2 + Timeline
D: Communications (i.e. to corresponding authors, mentors, and expert consultation, etc)
E: Draft of Materials and Methods (this can be performed for daily entries if variations occur over the course of the project).
F: Background- ie. competitor/market analysis, criteria/constraints
G: Daily Entries (every time you complete work on the project)
 1: Title and Date
 2: Short Introduction (putting the experiment/observations into context/objectives)
 3: Methods/Materials (if not included in the beginning of the notebook)
 Materials become important when someone needs to repeat your experiments
 4: Observations/Experimental Data (both RAW and ANALYZED)-
A: graphs/figures
B: data tables
C: pictures
D: sketches or proof of concepts and prototypes (with labels)
E: Decision matrices
E: Ethical responsibility
 5: Calculations and Data Analysis (STATISTICS)
 6: Final Concluding Remarks

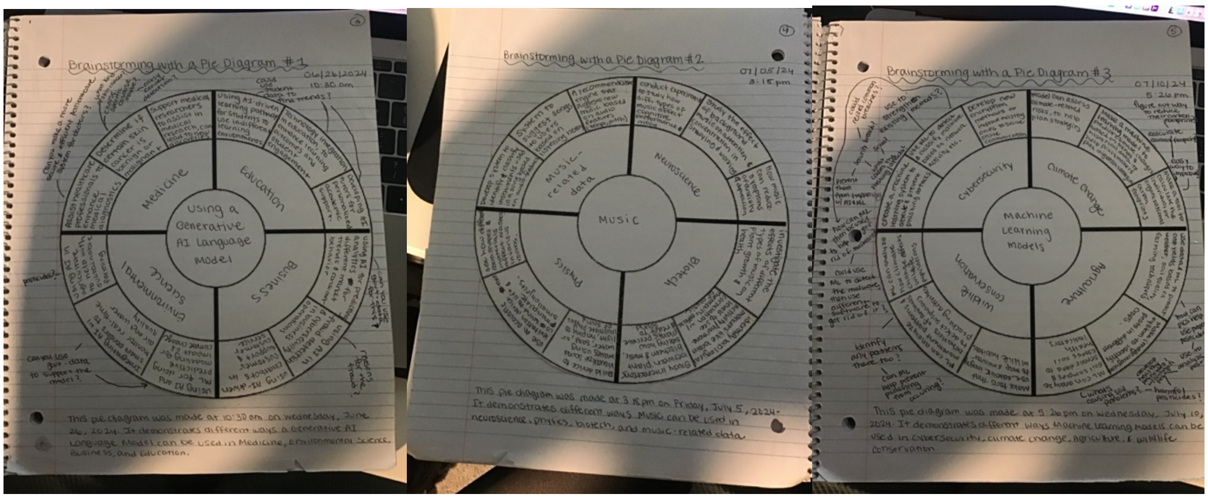
Things to keep in mind:

-You don't want to have too much blank space

-If you are adding a pre-printed graph or sketch, paste in and sign + date.

Brainstorming- provide a figure caption, sign and date any figures

Pie Diagrams:



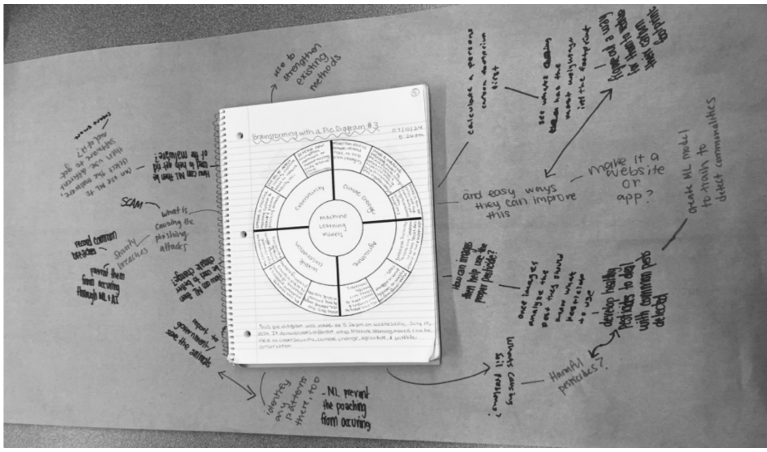
June 26, 2024 July 5, 2024 July 10, 2024

The first pie diagram talks about how Generative AI models can be used in Education, Medicine, Environmental Science, and Business.

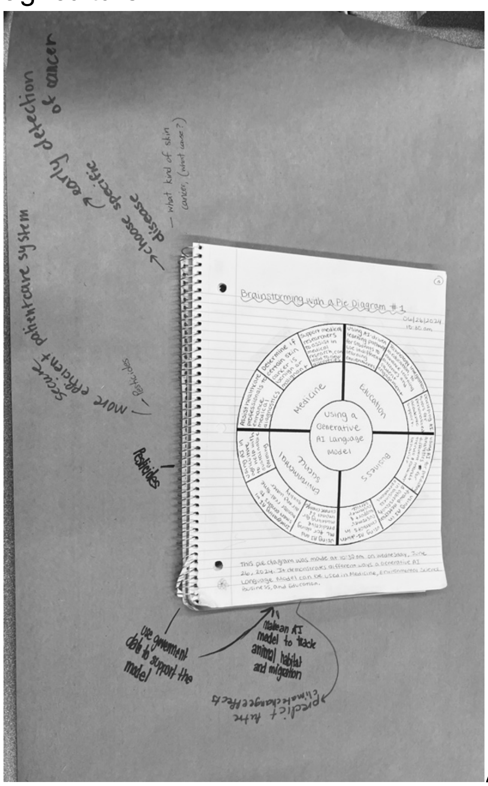
The second pie diagram talks about how Music can be combined with different scientific topics, like Physics, Biotech, Neuroscience, and Music-related data.

The third pie diagram talks about how Machine Learning model can be combined with cybersecurity, Climate change, wildlife conservation, and agriculture.

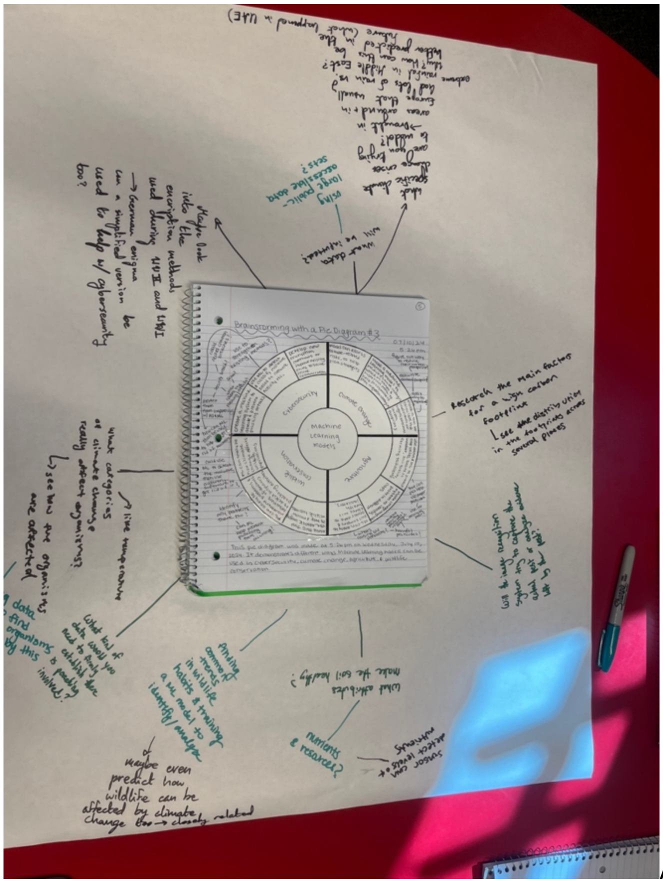
Mindmaps:


July 28th, 2024

This mind map shows future applications with Machine Learning models and their use in Cybersecurity, Wildlife conservation, but mostly focused on ideas for Climate change and agriculture.


August 11th, 2024

This mindmap mostly focused on using a generative AI model for environmental science and medicine.

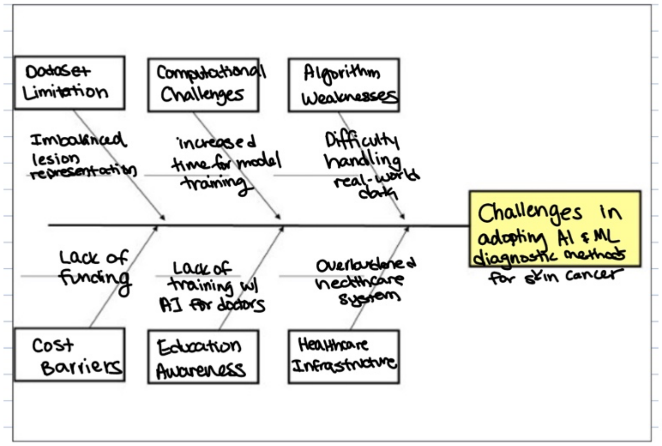


August 21st, 2024 

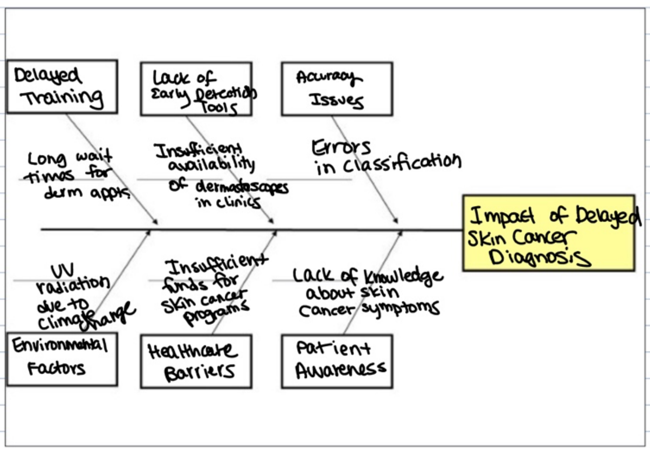
This mind map mostly shows how wildlife conservation can be integrated with a ML model, but also for climate change, agriculture, and cybersecurity.

Scamper

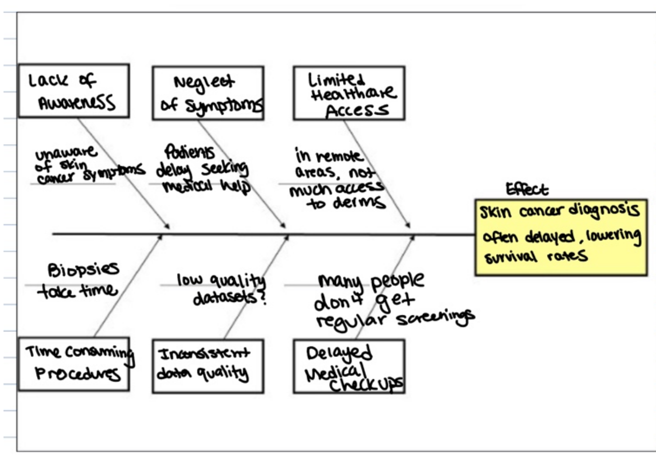
Fishbone Diagrams



11/24/24



11/24/24



11/24/24

5 Whys

Problem Statement: Skin cancer often goes undetected until it reaches an advanced stage, making treatment less effective.

Why 1: Why does skin cancer go undetected until advanced stages?
-> Because people often don't recognize early signs or symptoms of skin cancer.
Why 2: Why don't people recognize the signs of skin cancer?
-> Because they might lack awareness or education about what skin cancer looks like.
Why 3: Why is there a lack of education about skin cancer?
-> Because public health campaigns don't always reach everyone or emphasize prevention.
Why 4: Why don't public health campaigns reach everyone?
-> Because they may not be prioritized in all communities or funded in an adequate manner.
Why 5: Why is funding for skin cancer awareness not very adequate?
-> Because resources may be directed to other health issues seen as more urgent or widespread.

11/25/24

Problem Statement: Traditional diagnostic methods for skin cancer, like biopsies, are time-consuming and delay critical treatment.

Why 1: Why are traditional methods like biopsies time-consuming?
-> Because they involve multiple steps, including sample collection, lab testing, and result interpretation.
Why 2: Why do these steps take so much time?
-> Because labs often handle many samples, causing delays in processing.
Why 3: Why do labs have so many samples to process?
-> Because there's a high reliance on biopsies for skin cancer detection.
Why 4: Why is there such reliance on biopsies?
-> Because they are currently the most accurate method for diagnosing skin cancer.
Why 5: Why haven't faster methods replaced biopsies?
-> Because advanced diagnostic technologies, like AI or imaging, are still in development or not widely adopted.

11/25/24

Problem Statement: Delayed diagnosis of malignant skin cancer significantly lowers patient survival rates.

Why 1: Why does delayed diagnosis of malignant skin cancer lower survival rates?
-> Because treatment is more effective when started early in the disease's progression.
Why 2: Why is early treatment more effective?
-> Because it prevents the cancer from spreading to other parts of the body.
Why 3: Why does skin cancer spread to other parts of the body?
-> Because malignant cells multiply quickly and could possibly invade nearby tissues or organs.
Why 4: Why do malignant cells multiply quickly?
-> Because they are genetically programmed to grow and divide uncontrollably.
Why 5: Why is it hard to stop malignant cells from multiplying?
-> Because cancer cells often resist standard treatments like chemotherapy or radiation if started too late.

11/25/24

Project Introduction:

Define the Problem/Research Question/Mathematical Conjecture
State the Engineering Objective/Research Hypothesis/Mathematical Objective
Brief Overview
Engineering Design Brief or Research Outline

Professional Communication:

Email #1: (to sarslanay@wpi.edu)

Good afternoon,

I hope this email finds you well. My name is Anika Sivasankar, and I am a junior at the Massachusetts Academy of Math and Science at WPI. I am writing to you to seek your advice and guidance on a STEM project that I am passionate about pursuing for a 5 month long individual STEM Independent Research Project. My project idea involves training a Machine Learning model to differentiate between benign and malignant skin cancers in a more efficient way. I would also examine any pre-existing models to find any disadvantages that might need to be addressed in my model Given the importance of an early and accurate diagnosis of skin cancer in dermatology, I strongly believe that this project could contribute meaningfully to the field.

I read your journal article about “Distributed Musical Performances: Architecture and Stream Management.” One thing I personally found interesting about this article is how the project handles two quite different types of data streams: high-resolution audio and video, which need to be perfectly synced, and MIDI data from electronic instruments. It is interesting to see how they have developed a system that can manage both types of data, keeping everything in sync during a performance. This problem seems like something that is really challenging but fascinating in the data management field.

I would appreciate the opportunity to discuss my project idea further with you. Specifically, I am interested in your views on the feasibility of this project, any kinds of potential challenges, and any resources you would recommend as I begin this project, due to your expertise in the Computer Science and Data Sciences fields. I am open to any suggestions you might have on refining this project's scope as well.

I am eager to hear your thoughts about this project idea and would be grateful for any time you could spare to discuss this further. Please let me know if you are available for a meeting, or if there is a convenient time for us to talk.

Thank you very much for your time and consideration.

Best Regards,

Anika Sivasankar

asivasankar1@wpi.edu

Email #2: (to rneamtu@wpi.edu)

Good morning,

I hope this email finds you well. My name is Anika Sivasankar, and I am a junior at the Massachusetts Academy of Math and Science at WPI. I am writing to you to seek your advice and guidance on a STEM project that I am passionate about pursuing for a 5 month long individual STEM Independent Research Project. My project idea involves training a Machine Learning model to differentiate between benign and malignant skin cancers in a more efficient way. I would also examine any pre-existing models to find any disadvantages that might need to be addressed in my model Given the importance of an early and accurate diagnosis of skin cancer in dermatology, I strongly believe that this project could contribute meaningfully to the field.

I looked at your program at WPI, AI4ALL that studies ethical issues relating to AI, as well as the exploration of human impact of AI as a whole. I believe that with your strong approaches to AI from your perspective as a data mining researcher, as well as your expertise in finding techniques to explore data for use in various fields, such as medicine and neuroscience, your views could greatly help my project idea.

I would appreciate the opportunity to discuss my project idea further with you. Specifically, I am interested in your views on the feasibility of this project, any kinds of potential challenges, and any resources you would recommend as I begin this project, due to your expertise in the] Data Sciences fields. I am open to any suggestions you might have on refining this project's scope as well.

I am eager to hear your thoughts about this project idea and would be grateful for any time you could spare to discuss this further. Please let me know if you are available for a meeting, or if there is a convenient time for us to talk.

Thank you very much for your time and consideration.

Best Regards,

Anika Sivasankar

asivasankar1@wpi.edu

Exploring a STEM Project: Requesting for Your Input and Guidance

SA **Sivasankar, Anika <asivasankar1@wpi....** Wednesday, November 20, 2024 at 6:32 PM
To: Shue, Craig

Good evening,

I hope this email finds you well. My name is Anika Sivasankar, and I am a junior at the Massachusetts Academy of Math and Science at WPI. I am writing to you to seek your advice
and guidance on a STEM project that I am passionate about pursuing for a 5 month long individual STEM Independent Research Project. My project idea involves training a Machine Learning model to differentiate between benign and malignant skin cancers in a more efficient
way. I would also examine any pre-existing models to find any disadvantages that might need to be addressed in my model Given the importance of an early and accurate diagnosis of skin cancer in dermatology, I strongly believe that this project could contribute meaningfully to the field.

I would appreciate the opportunity to discuss my project idea further with you. Specifically, I am
interested in your views on the feasibility of this project, any kinds of potential challenges, and
any resources you would recommend as I begin this project, due to your expertise in the Computer Science and Data Sciences fields. I am open to any suggestions you might have on
refining this project's scope as well.

I am eager to hear your thoughts about this project idea and would be grateful for any time you
could spare to discuss this further. Please let me know if you are available for a meeting, or if
there is a convenient time for us to talk.

Thank you very much for your time and consideration.

SC **Shue, Craig <cshue@wpi.edu>** Thursday, November 21, 2024 at 2:46 PM
To: Sivasankar, Anika; Cc: Agu, Emmanuel

Hi Anika,

Thanks for reaching out!

I have cc'ed Prof. Emmanuel Agu. Prof. Agu has expertise in image analysis of wound healing associated with diabetes. You can see a list of his publications at his website below [1]. I do not have domain expertise in biology and could not tell you if image data is predictive for skin cancer. If it is, perhaps Prof. Agu's "Chronic Wound Image Augmentation and Assessment using Semi-Supervised Progressive Multi-Granularity EfficientNet" publication would be a good guide on tools and evaluation metrics you could use?

If Prof. Agu has time to chat, he is likely to be the best expert to engage. If he does not, perhaps he could recommend a discussion with one of his current graduate students to give pointers?

Best,

-- Craig

[1] https://web.cs.wpi.edu/~emmanuel/publications/index.html

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Craig A. Shue, Ph.D.
Professor and Department Head
Computer Science Department
Worcester Polytechnic Institute

SA Sivasankar, Anika <asivasankar1@wpi.edu> Friday, November 22, 2024 at 11:42 AM
To: Shue, Craig

Hi Dr. Shue,

Thank you so much for your advice! I will be in touch with Prof. Agu on my project and I will definitely read over that paper that you have referred me. I believe it will be a very useful resource.

Thank you once again!

Best,
Anika

AE Agu, Emmanuel <emmanuel@wpi.edu> Thursday, November 21, 2024 at 5:18 PM
To: Shue, Craig; Sivasankar, Anika

Hi Anika,

There's been quite a few papers in the area of skin cancer/melanoma and I think it's definitely feasible for AI-based image analses to distinguish benign vs. malignant cancers. My closes work to the work you describe discriminated Benign vs. malignant breast cancers:

https://www.mdpi.com/2227-9709/9/4/91

https://www.mdpi.com/2076-3417/13/1/156

You can start with a search on Google scholar (scholar.google.com/. The most recent models use neural networks. But for that you need lots Of data (a few hundred example benign and malignant images). You should Search for a public dataset, which would mean you don't have to collect your own data. I'm almost sure one exists.

If you know python, you can try to use python libraries such as scikit-learn for machine learning and pytorch for deep learning. But if you want an easier, non-programming route, you can also try libraries such as matlab that have drag and drop interfaces for machine learning.

Let me know if you have additional questions.

E

SA Sivasankar, Anika <asivasankar1@wpi.edu> Tuesday, November 26, 2024 at 4:46 PM
To: Agu, Emmanuel

Hi Professor Agu,

Thank you so much for your response and advice! I was wondering if we could have a Zoom meeting sometime this week to chat about this project, and so I could clear up some doubts I have regarding my project idea? Please let me know your availability.

Thank you!
Anika

Materials and Methods:

Daily Entries:

1/11/25 – Worked to see if model was performing properly and classifying all the images
1/15/25 – Started working on model that would take in image from user and diagnose
1/17/25 – Continued working on model that would take in image from user and diagnose
1/20/25 – Continued working on model that would take in image from user and diagnose
1/26/25 – Continued working on model that would take in image from user and diagnose
1/28/25 – Started working on ChatGPT Integration into model
1/29/25 – Continued working on ChatGPT Integration into model
2/2/25 – Continued working on ChatGPT Integration into model
2/5/25 – Continued working on ChatGPT Integration into model
2/6/25 – Continued working on ChatGPT Integration into model
2/7/25 – Continued working on ChatGPT Integration into model - tested it too
2/8/25 – Continued working on ChatGPT Integration into model - continuing testing

Project Time Management

Date	Activity Description	#hours	Approved
9/13/24	Reading about the role of AI in clinical Practices https://bmcmededuc.biomedcentral.com/articles/10.1186/s12909-023-04698-z	2 hours	
9/14/24	Reading about benefits of Machine Learning use in Healthcare https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8822225/	1.5 hours	
9/15/24	Read more form the article yesterday	1.5 hours	

9/17/24	Reading about the potential for AI to transform healthcare https://www.nature.com/articles/s41746-024-01097-6	1.5 hours	
9/18/24	Reading more about AI's role in healthcare https://www.mdpi.com/2075-1729/14/5/557	2 hours	
9/21/24	Continued reading article from the other day	1.5 hours	
9/23/24	Reading about bias and human trust in AI in healthcare https://www.jmir.org/2020/6/e15154/	1.5 hours	
9/24/24	Continued the article I read from yesterday	1 hour	
9/26/24	Reading about how AI can help in Cancer Research (early cancer detection through image analysis) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10697339/	2 hours	

9/27/24	Decided on an article for STEM Formal Meeting Presentation	1 hour	
9/30/24	Worked on STEM Formal Meeting Presentation	1 hour	
10/1/24	Worked on STEM Formal Meeting Presentation	2 hours	
10/2/24	Worked on revising and memorizing for STEM Formal Meeting	2 hours	
10/7/24	Did some research on Convolutional Neural Networks	2 hours	
10/8/24	More research on CNNs	1 hour	
10/9/24	Did some research on what kinds of image datasets would be best to train a ML model, based on CNNs	1 hour	
10/22/24	Worked on Build Something Project	2 hours	
10/23/24	Read some articles about CNNs	2 hours	
10/25/24	Worked on Build Something Project	1 hour	
10/29/24	Reading on Deep Learning Benefits	1 hour	
10/30/24	Watched Videos on how to create a base	2 hours	

	level CNN		
11/1/24	Watched Videos on how to create a base level CNN	1 hour	
11/4/24	Worked with mentor to begin a training model with dataset	2 hours	
11/6/24	Worked with mentor to begin a training model with dataset	2 hours	
11/7/24	Worked with mentor to begin a training model with dataset	1.5 hours	
11/12/24	Finding a new better Kaggle Dataset for training	1 hour	
11/14/24	Watching videos on CNN basics	2 hours	
11/16/24	Watching videos on CNN basics	2 hours	
11/19/24	Watching videos and reading articles on CNN basics	2 hours	
11/23/24	Worked with mentor on training model with new dataset	3 hours	
11/27/24	Worked on training images on the model		
11/29/24	Worked with mentor on model	3 hours	

12/4/24	Worked on Dec fair Poster	2 hours	
12/5/24	Worked on Dec fair Poster	2 hours	
12/8/24	Worked on testing model, with different images	2 hours	
12/10/24	Revised my poster with feedback from Dec Fair	2 hours	
12/12/24	Gathered my Prelim Data and made sure it was clear	1 hour	
12/14/24	Worked on slideshow for STEM formal meeting 2	2 hours	
1/6/25	Testing model to see if it could classify properly	1 hour	
1/7/25	Finding datasets with varying lighting	2 hours	
1/9/5	Finding datasets with varying skin tones	2 hours	
1/23/25	Working on model to take in an image	2 hours	
1/24/25	Working on model to take in an image	1 hour	
1/25/25	Working on model to take in an image	3 hours	

1/26/25	Working on model to take in an image	2 hours	
1/28/25	Started integrating ChatGPT with API	1 hour	
1/29/25	Coding ChatGPT into the model	2 hours	
2/2/25	Continued working on ChatGPT Integration into model	2 hours	
2/5/25	Continued working on ChatGPT Integration into model	2 hours	
2/6/25	Continued working on ChatGPT Integration into model	2 hours	
2/7/25	Tested the Model	3 hours	
2/8/25	Tested the Model	1 hour	